DDL Commands in MySQL – Exercise 1

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1. CREATE A DATABASE
   CREATE DATABASE CollegeDB;

   Output:

```
+--------------------+
| Database           |
+--------------------+
| bhavi              |
| collegedb          |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.020 sec)
```

2. USE A DATABASE
   USE CollegeDB;

   Output:

```
Database changed
```

3. CREATE A TABLE
   CREATE TABLE Students (StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT, Department VARCHAR(30), Marks INT);

   INSERT INTO Students VALUES (101, 'Arun', 20, 'CSE', 85), (102, 'Bala', 21, 'IT', 90), (103, 'Charan', 19, 'ECE', 78);

   Output:

```
+-----------+-------------+------+
| StudentID | StudentName | Age  |
+-----------+-------------+------+
|       101 | Arun        |   20 |
|       102 | Bala        |   21 |
|       103 | Charan      |   19 |
+-----------+-------------+------+
3 rows in set (0.007 sec)
```

4. DESCRIBE A TABLE

   DESC Students;

   Output:

```
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| StudentID   | int         | NO   | PRI | NULL    |       |
| StudentName | varchar(50) | YES  |     | NULL    |       |
| Age         | int         | YES  |     | NULL    |       |
| Department  | varchar(30) | YES  |     | NULL    |       |
| Marks       | int         | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.117 sec)
```

5. ADD A NEW COLUMN

   ALTER TABLE Students ADD Email VARCHAR(100);

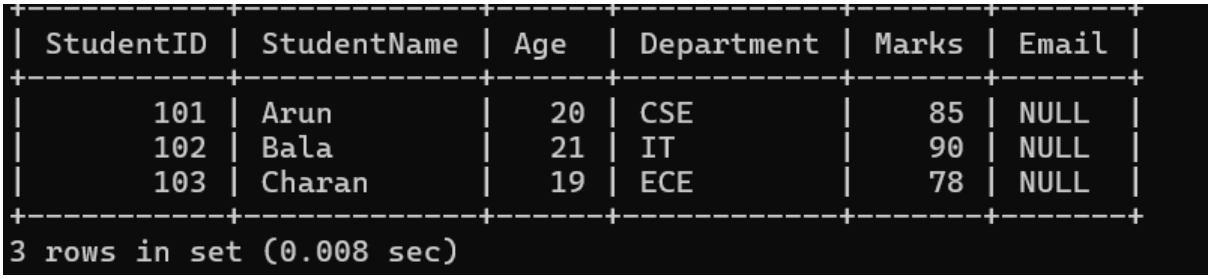
   Output:

```
+-----------+-------------+------+------------+-------+-------+
| StudentID | StudentName | Age  | Department | Marks | Email |
+-----------+-------------+------+------------+-------+-------+
|       101 | Arun        |   20 | CSE        |    85 | NULL  |
|       102 | Bala        |   21 | IT         |    90 | NULL  |
|       103 | Charan      |   19 | ECE        |    78 | NULL  |
+-----------+-------------+------+------------+-------+-------+
3 rows in set (0.008 sec)
```

6. MODIFY A COLUMN DATATYPE

   ALTER TABLE Students MODIFY StudentName VARCHAR(100);

   Output:



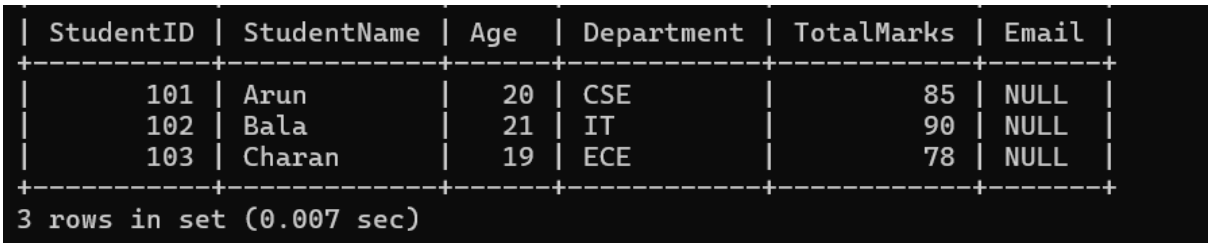
StudentID	StudentName	Age	Department	Marks	Email
101	Arun	20	CSE	85	NULL
102	Bala	21	IT	90	NULL
103	Charan	19	ECE	78	NULL

3 rows in set (0.008 sec)

7. RENAME A COLUMN

ALTER TABLE Students RENAME COLUMN Marks TO TotalMarks;

Output:



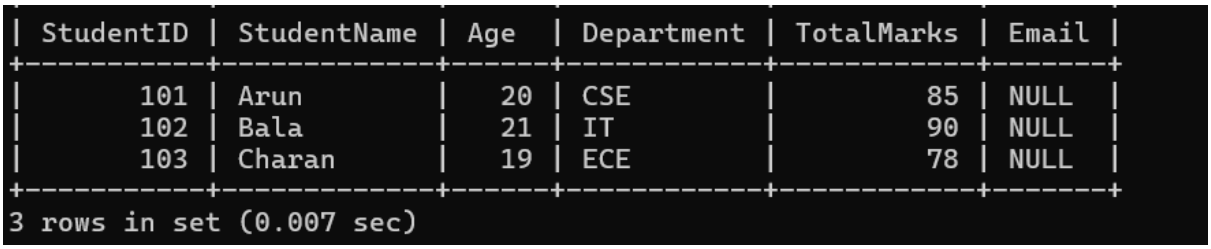
StudentID	StudentName	Age	Department	TotalMarks	Email
101	Arun	20	CSE	85	NULL
102	Bala	21	IT	90	NULL
103	Charan	19	ECE	78	NULL

3 rows in set (0.007 sec)

8. ADD A CONSTRAINT

ALTER TABLE Students ADD CONSTRAINT chk_age CHECK (Age >= 17);

Output:



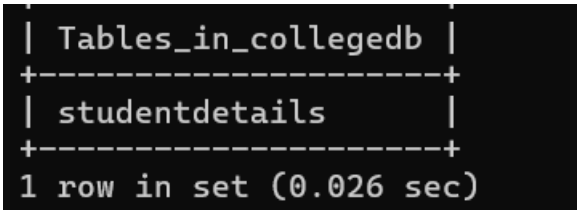
StudentID	StudentName	Age	Department	TotalMarks	Email
101	Arun	20	CSE	85	NULL
102	Bala	21	IT	90	NULL
103	Charan	19	ECE	78	NULL

3 rows in set (0.007 sec)

9. RENAME A TABLE

RENAME TABLE Students TO StudentDetails;

Output:



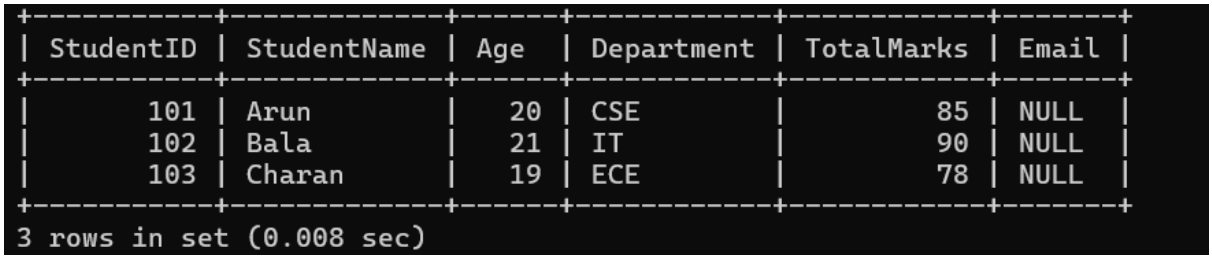
Tables_in_collegedb
studentdetails

1 row in set (0.026 sec)

10. CREATE INDEX

CREATE INDEX idx_department ON Students(Department);

Output:

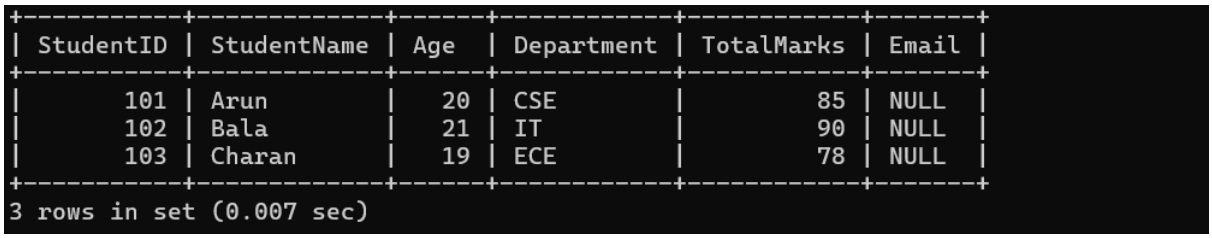


StudentID	StudentName	Age	Department	TotalMarks	Email
101	Arun	20	CSE	85	NULL
102	Bala	21	IT	90	NULL
103	Charan	19	ECE	78	NULL

3 rows in set (0.008 sec)

11. DROP INDEX
    DROP INDEX idx_department ON Students;

    Output:

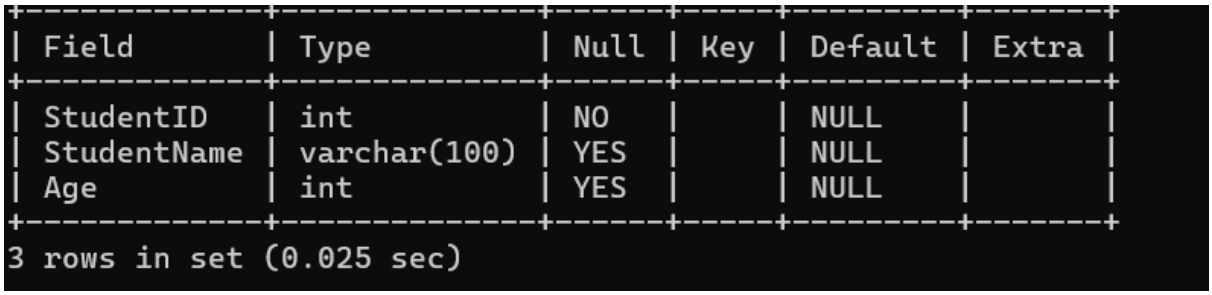


StudentID	StudentName	Age	Department	TotalMarks	Email
101	Arun	20	CSE	85	NULL
102	Bala	21	IT	90	NULL
103	Charan	19	ECE	78	NULL

3 rows in set (0.007 sec)

12. CREATE VIEW
    CREATE VIEW CSEStudents AS SELECT StudentID, StudentName, Age FROM Students WHERE Department = 'CSE';

    Output:



Field	Type	Null	Key	Default	Extra
StudentID	int	NO		NULL	
StudentName	varchar(100)	YES		NULL	
Age	int	YES		NULL	

3 rows in set (0.025 sec)

13. DROP VIEW
    DROP VIEW CSEStudents;

    Output:

```
mysql> DROP VIEW CSEStudents;
Query OK, 0 rows affected (0.104 sec)

mysql> desc CSEStudents;
ERROR 1146 (42S02): Table 'collegedb.csestudents' doesn't exist
```

14. DELETE A TABLE
    DROP TABLE Students;

    Output:

```
mysql> DROP TABLE Students;
Query OK, 0 rows affected (0.203 sec)

mysql> show tables;
Empty set (0.017 sec)
```

15. DELETE A DATABASE
    DROP DATABASE CollegeDB;

    Output:

```
mysql> DROP DATABASE CollegeDB;
Query OK, 0 rows affected (0.158 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| bhavi              |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.012 sec)
```